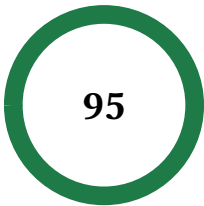


Recovery

fixture



95 out of 100

PRODUCTION READY

Gate: production-ready

Production ready under this rubric: Merge Readiness Score at least 85, measured coverage, and no critical gap. This is not a security audit.

Statements 100.0%	Branches 100.0%	Gaps 0	Mutation yes
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Score factors

Factor	Earned	Max	Detail
Structural coverage	40	40	Statement coverage 100.0% and branch coverage 100.0% combine to 100.0 before the 40-point weight.
Property-based resilience	25	25	Ran 349 property trials across structural conformance, determinism, and entity substitution. Failing inputs were shrunk toward simpler values.
Prompt / semantic alignment	10	15	ΔH coverage 66.7%. ΔH is the drop in Shannon entropy of specification tokens after removing tokens named by tests. This is a lexical spotlight, not transformer attention.

Blast radius safety	10	10	No diff was supplied. The radius is the whole indexed graph, so union coverage collapses to coverage of those functions.
Execution-time efficiency	10	10	Mann-Whitney U compared baseline inputs with heavier inputs on this same revision. A regression is recorded only when $p < 0.05$ and the median is more than 8x slower after allowing for a larger list input. This is not a cross-commit EffiBench run.

Coverage by module

Bars are drawn from the analysis. A gray bar is not a measured zero.

shop.cart		100%
shop.payments		100%
shop.pricing		100%

Gap severity

critical	0
high	0
medium	0
low	0

Files

File	Statements	Coverage	Gaps
src/shop/cart.py	19	100.0%	0
src/shop/payments.py	17	100.0%	0
src/shop/pricing.py	15	100.0%	0

Findings

No gaps recorded.

Property-based testing

Ran 349 property trials across structural conformance, determinism, and entity substitution. Failing inputs were shrunk toward simpler values.

Symbol	Trials	Passed	Failed
subtotal	191	191	0
apply_coupon	40	40	0
charge	34	34	0
refund	84	84	0

Mutation, prompt coverage, blast radius, timing

Killed 5/5 viable mutants on a temp copy (0 timeouts counted as killed, 0 unviable excluded). MSI 100.0.

MSI_total 100.0 keeps unviable mutants in the denominator. This is a sampled operator flip, not a mutmut or cosmic-ray campaign. Project files were not edited.

ΔH is the drop in Shannon entropy of specification tokens after removing tokens named by tests. This is a lexical spotlight, not transformer attention.

No diff was supplied. The radius is the whole indexed graph, so union coverage collapses to coverage of those functions.

Mann-Whitney U compared baseline inputs with heavier inputs on this same revision. A regression is recorded only when $p < 0.05$ and the median is more than 8x slower after allowing for a larger list input. This is not a cross-commit EffiBench run.

Authenticity scorecard

Authenticity scorecard is not the Merge Readiness Score and does not move the gate. DAR checks the workspace, the stdlib, and declared manifests. It does not query a package registry. FIRI is a static scan, not a reproduction of a flake.

Dimension	Value	Threshold	Status
Dependency authenticity (DAR)	100.0	100%	PASS
Statement coverage	100.0	$\geq 80\%$	PASS
Branch coverage	100.0	$\geq 75\%$	PASS
Mutation score (MSI)	100.0	$\geq 70\%$	PASS
Assertion strength (ASR)	100.0	$\geq 85\%$	PASS
Mean CRAP	5.6	≤ 15	PASS
CRAP > 30	0	0	PASS
Flakiness risk (FIRI)	0.0	0%	PASS

Suggestions

[- Read the findings from the top. Critical and high items are the ones that move the gate.] [- `recoverage generate` keeps a draft only after a temp-copy compile and 5 passing runs, and only if it adds covered lines or kills a mutant the current suite left alive.] [- Recoverage will not delete or overwrite an existing test file.] [- Prompt coverage is a lexical entropy proxy unless an attention model was queried. The report names which one ran.] [- Mutation counts come from temp-copy mutants. If that section says mutation did not run, it did not.] [- Re-run after editing tests: `recoverage run . --no-llm --threshold merge-ready`]

Rubric

Recoverage score rubric

The score is the Merge Readiness Score (MRS), 0–100. It is the PRD's weighted sum, not a line-coverage percentage.

Factor	Points	How it is measured
Structural coverage	40	60% project statement coverage + 40% branch coverage. Unmeasured runtime coverage scores 0. coverage.py's own `percent\_covered` is reported beside it and is not the score: with `--branch` that headline blends arcs into statements.
Property-based resilience	25	Share of property trials that held. The properties are structural conformance, determinism, and entity substitution. The engine runs at least 1,000 inputs when pure functions exist. Failing inputs are shrunk. This is not a frozen expected-output check.
Prompt / semantic alignment	15	ΔH , the drop in Shannon entropy of specification tokens once tokens named by tests are removed. Offline this is a lexical spotlight, not transformer attention. No spec text scores 0.
Blast radius safety	10	Union coverage of the call-graph radius. With no diff, the radius is the whole indexed graph and the report says so.
Execution-time efficiency	10	Mann-Whitney U between baseline inputs and heavier inputs on this revision. Points are lost only when $p < 0.05$ and the median is more than 8× slower. This is not a cross-commit benchmark.

Mutation testing is a separate measurement. When it runs, mutants are applied on a temp copy and the kill count is printed. It is not silently folded into MRS. If it does not run, the report says it did not run.

Flake markers found by a static scan subtract 2 points, floored at 0. That scan is not a reproduction. Drafts that `recoverage generate`

keeps are executed 5 times in a temp copy before they are copied back.

Gates

MRS is mapped onto four gates so a CI job can fail below a named bar. The PRD badge uses the same threshold, default 85.

Gate	Rule
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blocked	No test runner, or measured project statement coverage is below 20%, or MRS is below 40, or the test run did not exit 0. Badge: NOT MERGEABLE.
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needs-review	A runner exists and MRS is at least 40, but the merge bar is not met. Unmeasured runtime coverage cannot be graded above this gate.
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merge-ready	MRS $\geq$ 70, statement coverage $\geq$ 60%, runtime coverage was measured, and there is no critical gap. Badge: MERGEABLE.
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production-ready	MRS $\geq$ 85 (the PRD threshold), statement coverage $\geq$ 60%, runtime coverage was measured, and there is no critical gap. Badge: PRODUCTION READY.
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`--threshold` accepts a gate name or a number. A gate threshold passes only when the result gate ranks at least that high: blocked < needs-review < merge-ready < production-ready. A numeric threshold passes when MRS is greater than or equal to the number.

Separate from MRS

CRAP, assertion strength, dependency authenticity, static flakiness risk, and the mutation score are reported on an authenticity scorecard. They are not added into the 100 points and they do not move the gate. A generated draft is kept when it adds covered lines or kills a mutant the existing suite left alive. That filter is there because a coverage bump with a vacuous assertion is not evidence. Drafts still do not lock in an observed return value.